Stellar Dominion

A Cathedral of Light on the Moon

Projecting a Visible Emerald-Green Cross from the Moon's Northern Pole Visible Across the Entire Contiguous United States

> *For we are God's handiwork, created in Christ Jesus to do good works, which God prepared in advance for us to do.*
> — Ephesians 2:10 (NIV)

Design & Calculations

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April 17, 2026

Abstract

Stellar Dominion integrates a 3 kW–5 kW emerald-green fiber laser directly into a SpaceX Starship HLS that lands permanently on the lunar north pole. The Starship becomes the operational base: its 50 m stainless steel hull provides structure, thermal mass, and mounting area for SolAero IMM-α solar panels. A deployable boom extends the laser, telescope, and diffractive optical element from the top of the vehicle, pointing horizontally toward Earth. The cross fires every Sunday when the Moon is visible at night over the US, with aspirational first light on Easter Sunday, April 1, 2027. Total system mass is 1.4 t—a 39% reduction over a standalone pallet design. Core project cost is $42–114 million (projector, energy system, Starship modifications, and ground systems), with a Starship lunar cargo flight adding $20–50 million if purchased separately. Total lifecycle cost including launch and ten years of operations is $72–192 million.

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A Cathedral on the Moon

For as long as we have existed, humanity has looked up at the Moon and dreamed. Every civilization, every culture, every solitary soul on a dark night has felt the pull of that pale light. Now imagine changing what they see.

Standing anywhere in the continental United States on a clear night, you look up at the full Moon. And there—unmistakable, luminous, impossible to ignore—a brilliant **emerald-green cross** burns above the lunar disk. Not a projection on a building. Not a screen. A real structure, 384 400 km away, painting the sky with coherent light.

Stellar Dominion proposes building a cathedral on the Moon's northern pole—not a cathedral of worship, but of engineering. Inside it: a kilowatt-class laser and diffractive optics that project a cross-shaped beam across the Earth–Moon divide, blanketing all 48 contiguous states in a single, shared, naked-eye spectacle.

> **One beam. One Moon. 330 million people looking up at the same light.**

This is not science fiction. Every component—the laser, the optics, the power system, the launch vehicle—exists today or is in active commercial development. What is missing is the will to build something this audacious.

Mission Parameters at a Glance

Table 1: Mission parameters at a glance

Parameter	Value
Platform	SpaceX Starship HLS (permanent installation)
Location	Lunar north pole (peak of eternal light)
Target coverage	Contiguous United States (lower 48 states)
Cross pattern	0.7° total angular span, 532 nm green
Optical power	3 kW–5 kW
Power source	SolAero IMM-α, 50 m^2, 21.8 kW peak
System mass	1.4 t
Operation	Sundays only; first light Easter 2027
Design life	10 years
Core project cost (excl. launch)	$42–114 million
Total lifecycle cost (incl. launch + ops)	$72–192 million

System Architecture

The Stellar Dominion hardware integrates directly into a SpaceX Starship HLS that lands permanently on the lunar north pole. The Starship is not merely the delivery vehicle—it *is* the operational platform. No separate structure is required. All subsystems are either passive or software-controlled; no human presence is needed after commissioning.

What Starship Provides for Free

Table 2: Resources inherited from the Starship HLS platform

Resource	Benefit
50 m stainless steel hull	Structure, rigidity, mounting surface
Height	13 km horizon line-of-sight; clears local terrain
Thermal mass (>200 t steel)	Stabilises laser operating temperature
Power bus	Shared Starship HLS infrastructure
Comms	S-band/Ku-band supplemented by minimal custom avionics
Structural durability	Stainless steel in vacuum; effectively indefinite life

Stellar Dominion Subsystems

Table 3: Stellar Dominion subsystems and key specifications

Subsystem	Function	Key spec
Fiber laser	Generate green light	5 kW optical, 532 nm
Beam expander	Collimate to ~0.5 m aperture	$M^2 < 1.2$
DOE	Shape beam into cross pattern	0.7° span, 80% efficiency
Deployable boom	Position optics above hull	3–5 m carbon composite + gimbal
Solar array	Primary power	50 m^2 SolAero IMM-α, 20 kW
Battery bank	Eclipse bridging	325 kg Amprius Si-anode
Radiator panels	Reject laser waste heat	20 m^2, hull-mounted
Pointing assembly	Track Earth as Moon orbits	< 0.1° accuracy
Avionics + comms	Housekeeping, telemetry	Minimal custom; shares Starship

Optical Design

Laser

Green light at 532 nm sits at the peak of the human photopic response curve, maximizing perceived brightness per watt delivered. **Important clarification:** high-power fiber lasers such as the IPG YLS series operate at 1070 nm infrared. Reaching 532 nm requires a **second-harmonic generation (SHG)** crystal stage—a well-established nonlinear optical process, but one that requires custom engineering at the 5 kW CW level. CW SHG conversion efficiency is typically 60–80%, yielding a combined wall-plug efficiency of ≈25–35% (baseline: 30%). This is an engineering development task, not a COTS solution.

Table 4: Fiber laser parameters (with SHG stage)

Parameter	Value
Wavelength	532 nm (SHG from 1064 nm IR fiber laser)
Optical output	3 kW–5 kW CW
Wall-plug efficiency	25–35% (baseline: 30%, including SHG stage)
Electrical power	12 kW–20 kW
Beam quality	$M^2 < 1.2$
Rated lifetime	> 100 000 hours (≈ 11 years)
Mass (including SHG)	200 kg–300 kg

Radiometric Power Requirement

The cross must be bright enough to see with the naked eye against the night sky. We target an irradiance at Earth equivalent to a naked-eye magnitude ≈ 0 green source:

$$E_{\text{min}} \approx 1 \times 10^{-9}\ \text{W m}^{-2}$$

The cross consists of two arms, each 0.7° long and 0.05° thick. The solid angle of each arm is:

$$\begin{aligned}\Omega_{\text{arm}} &= \left(0.7^\circ \times \frac{\pi}{180^\circ}\right) \times \left(0.05^\circ \times \frac{\pi}{180^\circ}\right) \\ &= 1.06 \times 10^{-5}\ \text{sr}\end{aligned}$$

Total solid angle of the cross (two arms, minus small central overlap):

$$\Omega_{\text{cross}} = 2 \times 1.06 \times 10^{-5} \approx \mathbf{2.05 \times 10^{-5}}\ \text{sr}$$

Required optical power, using the inverse-square radiometric relation:

$$\boxed{P_{\text{opt}} = E \cdot \Omega \cdot d^2}$$

$$d^2 = \left(3.844 \times 10^8\ \text{m}\right)^2 = 1.477 \times 10^{17}\ \text{m}^2$$

$$P_{\text{opt}} = (10^{-9}) \cdot (2.05 \times 10^{-5}) \cdot (1.477 \times 10^{17}) \approx 3.0\,\text{kW}$$

With 5 kW deployed we operate with a comfortable 4.5 dB margin above threshold—allowing for optical losses, DOE efficiency, and atmospheric absorption.

Diffractive Optical Element

A diffractive optical element (DOE) is a thin phase-modulating optic that reshapes the incident Gaussian laser beam into an arbitrary far-field intensity pattern—in our case, a cross. The phase mask is computed using the **Gerchberg–Saxton** iterative Fourier algorithm [1], which alternately applies the target amplitude constraint in the far field and the physical constraint (unit amplitude) in the near field until convergence.

The DOE is a passive optical component: once manufactured, it requires no power, has no moving parts, and suffers no degradation in vacuum. A surface-relief fused-silica DOE achieves 80–95% diffraction efficiency.

The algorithm is already prototyped in this project. The DOE pattern generator is listed in Section 16.

Angular Geometry

Table 5: Earth–Moon angular geometry for US coverage

Quantity	Value
Earth–Moon distance d	384 400 km
Continental US east–west span	4500 km
US angular width θ_{US}	$\arctan\left(\frac{4500}{384400}\right) \approx 0.67^\circ$
Cross total span	0.7° (covers US with margin)
Cross arm thickness	0.05°
Moon apparent diameter	0.5°

Power System

Landing Site: Peaks of Eternal Light

The lunar north pole hosts several elevated ridges and crater rims that receive near-continuous sunlight throughout the lunar year—nicknamed "peaks of eternal light." The rim of Peary crater (88.6 °N) and adjacent ridgelines identified by NASA's Lunar Reconnaissance Orbiter (LRO) illumination maps receive sunlight 70–95% of the time, with eclipse periods of hours at most during lunar winter.

The solar constant at the Moon equals that at Earth (no atmosphere to attenuate): $G_\odot = 1361\ \text{W m}^{-2}$.

Solar Array: Rocket Lab (SolAero) IMM-α

Operating in vacuum with no weather or oxidation, we use **bare** IMM-α (inverted metamorphic multi-junction) cells on lightweight Kapton substrate — no cover glass required [9]. These cells are manufactured by Rocket Lab Space Systems, which acquired SolAero Technologies in 2022, and are commercially available through an established procurement path. This eliminates the largest single mass item in traditional space solar arrays.

Table 6: SolAero IMM-α solar array parameters

Parameter	Value
Technology	SolAero IMM-α, bare cells on Kapton
Cell efficiency	32% (AM0, beginning of life)
Areal density	$2\,\mathrm{kg\,m^{-2}}$ deployed
Deployment	Rollout panels unfurling from Starship hull

$$P_{\text{required}} \approx 20\,\mathrm{kW} \quad \text{(see power budget)} \tag{1}$$

$$\eta_{\text{cell}} = 0.32 \quad \text{(SolAero IMM-}\alpha\text{, BOL)} \tag{2}$$

$$\begin{aligned} P_{\text{array}} &= G_{\odot} \cdot \eta \cdot A \\ &= 1361 \times 0.32 \times A = 435\,A\,\mathrm{W\,m^{-2}} \end{aligned} \tag{3}$$

Setting $P_{\text{array}} \geq 20\,\mathrm{kW}$ (with degradation margin):

$$A = \frac{20\,000}{435} \approx 46.0\,\mathrm{m^2} \quad \rightarrow \quad \textbf{baseline: } 50\,\mathrm{m^2}\textbf{, } 100\,\mathrm{kg}$$

This compares to 500 kg for traditional rigid panels at $10\,\mathrm{kg\,m^{-2}}$—a **5× mass reduction**.

Upgrade path. Rocket Lab's next-generation **IMM-β** cell achieves 33.3% efficiency with improved specific power (W/kg). It is a drop-in replacement for IMM-α requiring no redesign of the array structure or power system. If available at mission procurement time, IMM-β would provide additional degradation margin at no mass penalty.

Battery Technology: Amprius Silicon-Anode

We baseline **Amprius SiCore** silicon-anode lithium-ion cells—the highest energy-density Li-ion cells commercially available. Amprius already ships to aerospace and defense customers [8].

Battery Sizing (Eclipse Bridging)

LRO illumination models for Peary crater rim show worst-case darkness periods of **up to 120–124 hours** (≈5 Earth days) during some lunar winters. The laser shuts down during eclipses (no observer sees an eclipsed Moon). Battery sustains avionics and thermal survival only, in two modes:

The system uses ephemeris to predict eclipse duration before entry and selects the appropriate mode. Battery capacity (unchanged):

$$\begin{aligned} E_{\text{capacity}} &= 90\,\mathrm{kW\,h} \\ E_{\text{usable}} &= 90 \times 0.80 = 72\,\mathrm{kW\,h} \end{aligned}$$

$$\text{At 500 W (hibernation):} \quad \frac{72\,000}{500} = 144\,\mathrm{h} > 124\,\mathrm{h} \checkmark$$

At $280\,\mathrm{W\,h\,kg^{-1}}$ pack-level (Amprius):

$$m_{\text{battery}} = \frac{90\,000}{280} = 321\,\mathrm{kg} \quad \rightarrow \quad \textbf{baseline: } 325\,\mathrm{kg}$$

Power Budget

Thermal Management

Waste Heat

The laser subsystem (IR source + SHG stage) produces 12 kW of waste heat (electrical input minus optical output). Combined with avionics, total heat rejection is:

$$\begin{aligned} Q_{\text{total}} &= P_{\text{elec}} - P_{\text{opt}} + P_{\text{avionics}} \\ &\approx 17\,000 - 5\,000 + 500 = 12.5\,\mathrm{kW} \end{aligned}$$

Radiator Sizing

In lunar vacuum, heat can only be rejected by radiation. From the Stefan–Boltzmann law:

$$Q = \varepsilon \cdot \sigma \cdot A \cdot \left(T_{\text{rad}}^4 - T_{\text{env}}^4\right)$$

where $\varepsilon = 0.85$ (black anodized aluminum), $\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$, $T_{\text{rad}} = 340\,\mathrm{K}$ (laser operating temp), and $T_{\text{env}} = 100\,\mathrm{K}$ (polar sky background, conservative):

$$\frac{Q}{A} = 0.85 \times 5.67 \times 10^{-8} \times \left(340^4 - 100^4\right) \approx 640\,\mathrm{W\,m^{-2}}$$

$$A_{\text{radiator}} = \frac{12\,500}{640} \approx 19.5\,\mathrm{m^2} \quad \rightarrow \quad \textbf{baseline: } 20\,\mathrm{m^2}$$

Mass Budget

At 1.4 t—a 39% reduction over the standalone pallet architecture—the Stellar Dominion hardware is a small fraction of Starship HLS delivery capacity. The mass savings stem from eliminating the pallet structure entirely (400 kg) and from SolAero's 5× lighter solar panels (100 kg vs. 500 kg).

Table 7: Amprius SiCore battery energy density: cell vs. pack

Level	Energy density	Notes
Cell	400 W h kg^{-1}	Amprius SiCore, shipping
Pack (custom vacuum-rated)	~280 W h kg^{-1}	70% cell-to-pack; radiative cooling, composite enclosure, BMS

Table 8: Eclipse-mode power loads

Mode	Load	Conditions
Full survival	~1 kW	Short eclipses (≤72 hr): full avionics + heaters
Hibernation	~500 W	Extended eclipses (>72 hr): battery heat + avionics keep-alive

Landing, Deployment, and Pointing

Landing Site

Target: rim of Peary crater (88.6 °N) or an adjacent ridgeline with confirmed > 70% solar illumination per LRO illumination maps. Starship HLS employs terrain-relative navigation (TRN) for metre-scale landing accuracy.

Starship stays permanently. After touchdown, the engines shut down and the vehicle becomes the operational platform. There is no departure.

Height advantage. With the laser system at ~50 m (near the top of Starship), the geometric line-of-sight to the horizon is:

$$d_{\text{horizon}} = \sqrt{2R_{\text{Moon}}h} = \sqrt{2 \times 1\,737\,400 \times 50} \approx 13.2\,\text{km}$$

This comfortably clears local terrain, reducing sensitivity to the exact landing position within the polar region.

Deployment Sequence

1. Starship lands at polar peak (pre-surveyed via LRO); engines shut down permanently
2. SolAero panels unfurl from hull exterior; battery begins charging
3. System boots from Starship power bus; self-check; Earth comm link established
4. Star tracker acquires attitude; JPL ephemeris loaded via uplink
5. Deployable boom extends from nosecone area; laser/optics at operational position
6. Laser fires at 10% power; Earth ground station confirms pointing
7. Full-power commissioning; cross pattern confirmed
8. **First Light** — system enters autonomous Sunday operation schedule

Estimated commissioning time: 2–4 weeks post-landing.

Long-Term Structural Durability

Starship's 304L stainless steel hull is well-suited to permanent lunar deployment: no corrosion (no oxygen or moisture), mild polar thermal cycling (~50 K vs. 300 K at equatorial sites), and no wind, seismic, or weather loading. Structural design life exceeds the 10-year mission life by a large margin.

Lunar Surface Environment

Landing Plume and Dust

Starship's descent engines displace loose regolith during landing. In vacuum, displaced particles follow **ballistic trajectories**—there is no lingering cloud (no atmosphere to suspend particles), and ejecta settles within seconds of engine shutdown. All sensitive surfaces (solar panels, optics, DOE) are stowed in protective enclosures during landing and deployed only after the plume has fully dispersed.

Once deployed, there is **no ongoing aeolian transport**—without an atmosphere, surface material cannot be re-mobilized by wind. Electrostatic dust levitation exists near the terminator but produces negligible mass flux at 50 m elevation. Anti-static surface coatings on optical elements are a prudent precaution. Micrometeorite-driven regolith gardening produces a very low secondary ejecta flux over 10 years—relevant for surface abrasion studies but not a mission-level concern for fused-silica optics.

Radiation Environment

The Moon has no magnetosphere. The system is continuously exposed to galactic cosmic rays (GCR) and sporadic solar particle events (SPE). Mitigations:

- Starship's ~4 mm stainless steel hull provides ≈5–10 g/cm^2 of areal shielding for internally mounted

Table 9: Steady-state power budget

Subsystem	Power (W)	Notes
Fiber laser @ 30% efficiency (incl. SHG)	17 000	Dominant load
Pointing assembly	200	Low-power servo drives
Avionics + communications	300	Housekeeping, telemetry
Thermal heaters	500	Laser pre-heating during cold periods
10% margin	1 800	
Total	**19 800**	≈ 20 kW

Table 10: System mass budget

Subsystem	Mass (kg)	Basis
Fiber laser assembly (incl. SHG)	280	IPG YLS class + SHG crystal assembly, space-qualified
DOE + beam optics	200	0.5 m aperture, lightweight mirror
Solar array (50 m^2, SolAero IMM-α)	100	2 kg/m^2, bare cells on Kapton
Battery bank (Amprius Si-anode)	325	90 kWh @ 280 Wh/kg pack
Radiator panels (20 m^2, hull-mounted)	100	Flush on Starship hull
Pointing assembly	100	Star tracker + 2-axis gimbal
Avionics + communications	40	Minimal custom; shares Starship comms
Deployable boom (3–5 m)	75	Carbon composite + gimbal interface
Structure/pallet	*0*	*Eliminated — Starship IS the structure*
15% integration margin	183	
Total	**1 403**	**≈ 1.4 tonnes**

electronics, attenuating the lower-energy SPE proton spectrum.

- Laser diodes and SHG crystal are inside the boom enclosure with additional local shielding.
- Sunday-only operation limits cumulative dose to active electronics (≈416 operating hours/year vs. 8 760 for continuous operation).
- Avionics baseline: radiation-hardened design (latch-up tolerant, EDAC on memory)—standard for lunar surface missions.

Open item: A dedicated radiation transport analysis (SHIELDOSE or equivalent) is needed to quantify worst-case SPE fluence and size any additional local shielding for the avionics enclosure.

Pointing and Tracking

The laser must track the continental United States (38 °N, 97 °W) as the Moon orbits Earth and Earth rotates.

Earth is always near the horizon. From the lunar north pole, Earth sits at 0 ° elevation (±6.7° with libration in latitude). The boom-mounted gimbal therefore points roughly horizontally—a geometrically simple configuration that clears the vehicle body without requiring an overhead slew. As Earth drifts slowly across the horizon driven by libration, the gimbal compensates with a gentle elevation adjustment well within its travel range.

Visibility Analysis

Conditions for Visibility

Three conditions must be simultaneously satisfied for a US observer to see the cross:

1. The Moon is **above the horizon** at the observer's location
2. It is **night** (astronomical twilight or darker)
3. The Moon is in a **gibbous or full phase**—sufficient illumination for the Moon to be a landmark, and sky dark enough for the cross to pop

Lunar Libration and North Pole Visibility

The laser sits at the Moon's north pole. From Earth, the north pole is visible at an angle that varies with *libration in latitude*: the Moon nods ±6.7° over its ≈ 27-day orbit. When libration tilts the pole toward Earth, the cross projector has maximum line-of-sight exposure. Even when tilted away, the beam geometry is unchanged—the cross still covers the US.

Table 11: Pointing and tracking hardware

Component	Spec	Heritage
Star tracker	$< 5''$ accuracy	Jena-Optronik Astro APS class
2-axis gimbal	$0.004°$/s max slew rate needed	Commercial industrial
Ephemeris	JPL DE430	Pre-loaded + uplink corrections
Update rate	1 Hz	Earth moves $0.004°$/s in lunar sky
Required accuracy	$< 0.1°$	Cross arm is $0.7°$ wide

Monthly Visible Hours (Mid-Latitude US Observer)

The cross is most spectacular at full Moon—when all 330 million Americans can look up simultaneously and see the same light.

Sunday Operation Mode

The baseline design supports continuous autonomous operation, but the cross fires **only on Sundays**—when the Moon is visible at night over the United States. This is a scheduling decision, not a hardware redesign: the same 50 m^2 SolAero array and 325 kg Amprius battery are retained.

Why Not Shrink the Hardware?

One might propose a smaller solar array (~7 m^2) that charges a larger battery all week, discharging on Sunday. With SolAero panels at 2 kg m^{-2}, the baseline wins decisively:
The baseline is **264 kg lighter**—the opposite of a wash. SolAero's low areal density makes large arrays cheap on mass, so the week-long charging strategy loses its mass incentive entirely. It also adds single-point failure risk and deep cycling stress. The baseline hardware is kept as-is.

Benefits of Sunday-Only Operation

Cost Estimates

All figures in 2025 USD. Ranges reflect uncertainty between COTS and space-qualified component pricing (typically 5×–20× premium for space qualification).
The core project—the projector, energy system, and Starship modifications—costs **$42–114M**. Launch is a separable line item: if the hardware rides to the Moon as manifested cargo on a Starship flight that is flying anyway, the incremental launch cost is zero. Even paying the full dedicated flight, the complete mission fits within the budget of a medium-class space mission.

For perspective: the Hubble Space Telescope cost approximately $1.5 billion in 1990 dollars. A single NASA CLPS lunar lander contract costs $100–400 M. Stellar Dominion is achievable for the cost of one CLPS contract—and would be seen by orders of magnitude more people.

Operational Lifetime and Costs

Component Lifetimes

Mission design life: 10 years.

Annual Operating Costs (Sunday Operation Mode)

Ten-year operating cost: approximately $10–28 M.

Core project + 10 years ops (no launch): $52–142 million.
Full lifecycle (core + Starship flight + 10 years ops): $72–192 million.

Project Timeline

Aspirational first light: Easter Sunday, April 1, 2027.

Working backward: first light on Easter requires launch by early March 2027, payload integration by mid-February, and a 10-month compressed build starting April 2026. Compression is achieved by ordering the laser at kickoff, overlapping integration with late hardware deliveries, and running thermal-vacuum and vibration tests in parallel.
Schedule compression premium: +$8–20M (overtime, parallel streams, expedited shipping, higher contingency).

Hard dependency: Starship HLS availability. The gating item is not our build but whether SpaceX is flying lunar cargo by early 2027. If Starship is not available, the Easter 2027 target is not achievable regardless of our readiness. The 10-month build ensures we are ready the moment Starship is.

Regulatory Notes

Outer Space Treaty (1967), Article II
Prohibits *national* appropriation of the Moon. Private installations for peaceful purposes are not prohibited;

Table 12: Monthly visible viewing hours for a mid-latitude US observer

Phase	Sky condition	Hours/night	Nights/month
Full Moon	Best; brilliant cross on dark sky	~6	~3
Waxing/waning gibbous	Good	4–8	~8
Quarter	Marginal (competing twilight glow)	~4	~4
Crescent	Poor	<2	few
Useful viewing hours per month			**≈ 30–50 hours**

Table 13: Baseline vs. small-solar/burst-battery mass comparison

Architecture	Solar (kg)	Battery (kg)	Total (kg)
Baseline (50 m^2 SolAero + survival battery)	100	325	425
Small solar + burst battery (7 m^2 SolAero)	14	675	689

Table 14: Benefits of Sunday-only operation

Benefit	Impact
Laser lifetime	~416 hrs/year vs. 8,760 → effectively indefinite (>240 years at rated 100,000 hrs)
Thermal cycling	Reduced stress on optics and laser diodes
Ground operations	Sunday-only monitoring saves $5–10M over 10 years
Narrative resonance	The cross appears on the Sabbath; Easter Sunday is the natural inaugural event

the 2015 US SPACE Act (51 U.S.C. § 51303) confirms US entities may own hardware on celestial bodies.

Laser safety
At Earth, the cross irradiance ($\approx 1\times10^{-9}$ W m^{-2}) is orders of magnitude below the ANSI Z136.1 Maximum Permissible Exposure for extended-source viewing. The beam is fully eye-safe for diffuse observation.

Aviation
FAA NOTAMs and precision ephemeris-based pointing ensure the beam does not intersect aircraft corridors. The beam enters from ~ 45° elevation or higher—well above Class A airspace in a narrow geometric cone.

Export control
Multi-kilowatt lasers are ITAR/EAR-controlled items. Standard export licensing applies; no fundamental obstacles for a US-based program.

Risks and Open Issues

Visualization

The following Python script generates a scientifically accurate sky diagram of the cross as it would appear to a US observer: the Moon at its exact 0.5° apparent diameter, with the 0.7° cross centered at the northern limb, rendered at true angular proportions.

Sky Visualization Code

DOE Pattern Generator

The DOE phase pattern is computed using the Gerchberg–Saxton (GS) iterative Fourier algorithm [1]. The algorithm alternates between the spatial domain (imposing unit amplitude at the DOE plane) and the Fourier domain (imposing the target cross amplitude at the focal plane) until convergence. Output is a grayscale phase mask suitable for printing on transparency film (tabletop demo) or fabrication as a surface-relief fused-silica optic (flight unit).

Why Build It?

Because some things deserve to exist simply for the awe they inspire.

Stellar Dominion would be the first human-made structure visible from Earth with the naked eye beyond low orbit. It would be proof—silent, luminous, undeniable—that we are a species capable of reaching beyond our world and leaving a mark of beauty on another.

> Every child who looks up and sees it will know: *we built that.* Every scientist, artist, and dreamer

Table 15: Core project and full lifecycle cost estimates (2025 USD)

Item	Low ($M)	High ($M)	Basis
Core Project — projector, energy system, Starship modifications			
Laser system (5 kW, space-qual)	5	15	$200–500/W; 5–20× space premium
DOE + beam optics	2	5	Custom fused-silica DOE + telescope
Solar array (50 m^2, SolAero IMM-α)	5	12	Premium cells; less area than traditional
Battery bank (Amprius Si-anode)	3	8	325 kg; COTS cells + custom space pack
Radiators + thermal hardware	1	2	Hull-mounted; standard spacecraft thermal
Pointing assembly + boom	3	8	Star tracker, 2-axis gimbal, carbon boom
Avionics + communications	1	3	Minimal custom; shares Starship S-band
Starship modifications + integration	3	8	Hull/power/comms interfaces with SpaceX
System integration & test	8	20	Thermal-vac, vibration, EMI
Ground station setup	3	8	Antenna, control software
Project management + contingency	8	25	15–20% of hardware cost
Core project subtotal	*42*	*114*	*without launch*
Launch & Delivery			
Starship lunar cargo flight	20	50	SpaceX projected lunar cargo pricing
Grand total	**62**	**164**	**≈ $60–165M**

Table 16: Component rated lifetimes

Component	Rated Life	Notes
Fiber laser	>100 000 hours (≈11 years)	Sunday-only: ~416 hrs/yr → indefinite
Solar array	15–20 years	1–3%/year radiation degradation
Amprius Si-anode battery	1 000–2 000 cycles	Sunday-only cycling extends life well beyond 10 years
Avionics	10–15 years	Radiation-hardened design
DOE (passive optic)	Indefinite	No moving parts; vacuum-stable
Pointing gimbal	7–10 years	Wear item; oversize for life margin

Table 17: Annual operating costs (Sunday-mode)

Item	Low ($/yr)	High ($/yr)
Ground station ops (Sunday + weekly health)	500 000	1 500 000
Mission operations team (1–3 FTE)	300 000	800 000
Ephemeris updates + uplink	50 000	200 000
Contingency	150 000	300 000
Annual total	**1 000 000**	**2 800 000**

> will know it too. Not a flag planted in dust and forgotten. A light that shines back, night after night, for generations.

The engineering is feasible. The physics is settled. The components exist. The launch vehicle is being built. The cost is within the reach of a determined private effort or a modest government investment.

The question is no longer *can we*—it is *will we*.

References

References

[1] R. W. Gerchberg and W. O. Saxton, "A practical algorithm for the determination of phase from image and diffraction plane pictures," *Optik*, vol. 35, pp. 237–246, 1972.

[2] NASA Goddard Space Flight Center, "Lunar Fact Sheet," https://nssdc.gsfc.nasa.gov/planetary/factsheet/moonfact.html

[3] NASA Lunar Reconnaissance Orbiter Camera (LROC) team, "Illumination maps of the lunar polar regions," GSFC.

[4] IPG Photonics Corporation, "YLS Series High Power CW Fiber Lasers," product literature.

Table 18: Compressed 10-month project schedule to Easter 2027 first light

Month	Milestone	Deliverables
0	Kickoff + laser order	Requirements baseline; laser ordered immediately
1.5	PDR	Architecture frozen
2	DOE fabrication begins	GS pattern → fused-silica vendor
2–5	Solar + battery delivery	Amprius cells + pack; space-grade solar
4	CDR	Detailed design complete
5	Integration begins	Overlaps with late deliveries
6–8	Env. testing (parallel)	Thermal-vac + vibration campaign
8	Starship payload integration	Subsystems installed and verified in Starship
9	Launch readiness review	All tests passed
10	**Launch-ready**	**Ready for Starship lunar cargo flight**
L+30 d	**First Light**	**Easter Sunday, April 1, 2027**

Table 19: Risk register

Risk	Severity	Mitigation	Status
Starship landing reliability	High	SpaceX flight heritage; gating dependency	External dep.
Landing plume contamination	Medium	Stow during landing; deploy after	Addressed
Radiation (GCR/SPE)	Medium	Hull shielding; rad-hard avionics; Sunday-only limits dose	Analysis needed
Deep eclipse >72 hr	Low	500 W hibernation; 90 kWh covers 144 hr	Addressed
5 kW CW SHG at scale	Medium	Physics proven; engineering dev. required	Dev. risk
DOE degradation	Low	Fused-silica UV-stable; low micrometeorite flux	Long-term test needed
IMM-α availability	Low	Rocket Lab commercial supplier; 50 m^2 feasible	Monitor lead times
First-light verification	Low	US ground stations + citizen-science network	Ops planning

[5] R. W. Boyd, *Radiometry and the Detection of Optical Radiation*. Wiley-Interscience, 1983.

[6] American National Standards Institute, "ANSI Z136.1: Safe Use of Lasers" and "ANSI Z136.6: Safe Use of Lasers Outdoors."

[7] United Nations Office for Outer Space Affairs, "Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space," 1967.

[8] Amprius Technologies, "SiCore Silicon-Anode Lithium-Ion Cells," product data sheet, 400 W h kg^{-1} cell-level energy density.

[9] Rocket Lab Space Systems (formerly SolAero Technologies), "IMM-α and IMM-β Inverted Metamorphic Multi-Junction Solar Cells," product data; IMM-α: 33% AM0 efficiency, 2 kg m^{-2} bare cells on Kapton; IMM-β: 33.3% with improved specific power. https://www.rocketlabusa.com/space-systems/solar/

[10] U.S. Congress, "U.S. Commercial Space Launch Competitiveness Act," 51 U.S.C. § 51303, 2015.

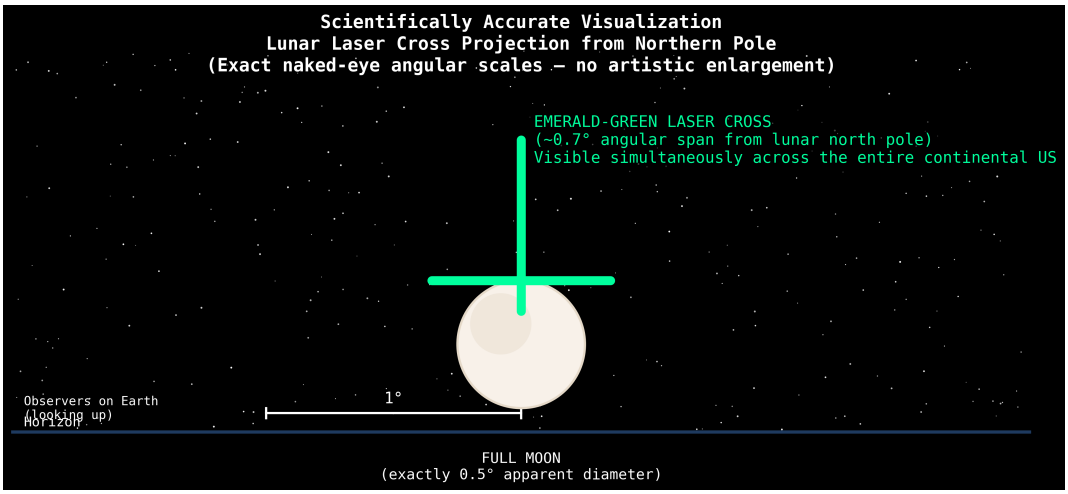

Figure 1: Scientifically accurate sky visualization. The Moon (0.5° apparent diameter) is shown at true angular scale. The emerald-green cross (0.7° total span, 0.05° arm thickness) originates at the Moon's northern limb. A 1° scale bar is shown for reference.

```python
"""Scientifically accurate lunar laser cross visualization at exact naked-eye angular scales."""
# pylint: disable=invalid-name
import matplotlib.pyplot as plt
import numpy as np
from matplotlib.patches import Circle

# ================================================
# Scientifically Accurate Lunar Laser Cross Visualization
# Exact naked-eye angular scales (no artistic enlargement)
# ================================================

fig = plt.figure(figsize=(14, 9), facecolor='black')
ax = fig.add_subplot(111)
ax.set_facecolor('black')

# Set angular coordinate system (degrees on sky)
# x = horizontal angular position (degrees)
# y = vertical angular position (degrees above horizon)
ax.set_xlim(-2.0, 2.0)
ax.set_ylim(0.0, 1.5)          # Moon low at ~15° but scaled for visual clarity
ax.set_aspect('equal')         # True angular proportions

# Remove axes spines and ticks for clean sky look
ax.axis('off')

# ==================== STARS ====================
np.random.seed(42)  # reproducible star field
N_STARS = 300
star_x = np.random.uniform(-2.0, 2.0, N_STARS)
star_y = np.random.uniform(0.0, 1.5, N_STARS)
ax.scatter(star_x, star_y, color='white', s=np.random.uniform(0.5, 3, N_STARS),
           alpha=np.random.uniform(0.6, 1.0, N_STARS), edgecolor='none')

# ==================== HORIZON ====================
ax.axhline(y=0.0, color='#1e3a5f', linewidth=6, zorder=1, label='Horizon')
ax.text(-1.95, 0.03, 'Horizon', color='white', fontsize=12, fontfamily='monospace')

# ==================== SCALE BAR (1°) ====================
ax.plot([-1.0, 0.0], [0.08, 0.08], color='white', linewidth=2)
ax.plot([-1.0, -1.0], [0.06, 0.10], color='white', linewidth=2)
ax.plot([0.0, 0.0], [0.06, 0.10], color='white', linewidth=2)
ax.text(-0.5, 0.12, '1°', color='white', fontsize=14, ha='center', fontfamily='monospace')
ax.text(-1.95, 0.15, 'Observers on Earth\n(looking up)', color='white', fontsize=11,
        ha='left', va='top', fontfamily='monospace')

# ==================== FULL MOON (exactly 0.5° diameter) ====================
MOON_CENTER_X = 0.0
MOON_CENTER_Y = 0.35
MOON_RADIUS = 0.25   # = 0.5° diameter

moon = Circle((MOON_CENTER_X, MOON_CENTER_Y), MOON_RADIUS,
              facecolor='#f8f1e9', edgecolor='#e5d9c8', linewidth=2, zorder=10)
ax.add_patch(moon)

# Slight lunar surface shading for realism
ax.add_patch(Circle((MOON_CENTER_X - 0.08, MOON_CENTER_Y + 0.08), 0.12,
                    facecolor='#e5d9c8', alpha=0.4, zorder=11))

# Label for Moon
ax.text(MOON_CENTER_X, MOON_CENTER_Y - 0.42, 'FULL MOON\n(exactly 0.5° apparent diameter)',
        color='white', fontsize=13, ha='center', va='top', fontfamily='monospace')

# ==================== EMERALD-GREEN LASER CROSS ====================
# Center of cross = lunar northern limb (top of Moon)
CROSS_CENTER_X = MOON_CENTER_X
CROSS_CENTER_Y = MOON_CENTER_Y + MOON_RADIUS   # exactly at top limb

# Cross total span = 0.7° (matches continental US angular width from Moon)
# Each arm: 0.7° long × 0.05° thick (matches README solid-angle calculation)
CROSS_HALF_SPAN = 0.35         # half of 0.7°
CROSS_HALF_THICKNESS = 0.025  # half of 0.05°

# Horizontal arm (filled rectangle in data coordinates)
```

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.fftpack import fft2, ifft2

N = 2048                        # high-res for sharp printing
np.random.seed(42)

# Target cross (slightly thicker arms for visible room demo)
target = np.zeros((N, N))
arm_thickness = int(N * 0.08)
center = N // 2
target[center - arm_thickness//2 : center + arm_thickness//2, :] = 1.0
target[:, center - arm_thickness//2 : center + arm_thickness//2] = 1.0
target /= np.sum(target)

# Gerchberg-Saxton algorithm
amplitude_target = np.sqrt(target)
phase = 2 * np.pi * np.random.rand(N, N)
field = amplitude_target * np.exp(1j * phase)

for _ in range(8):          # more iterations = cleaner cross
    input_field = ifft2(field)
    input_phase = np.angle(input_field)
    input_complex = np.exp(1j * input_phase)
    field = fft2(input_complex)
    field = amplitude_target * np.exp(1j * np.angle(field))

doe_phase = np.angle(input_complex) % (2 * np.pi)

# Printable versions
grayscale_doe = (doe_phase / (2 * np.pi) * 255).astype(np.uint8)
binary_doe = np.where(grayscale_doe > 127, 255, 0)

# Save for printing
plt.imsave('doe_grayscale_print.png', grayscale_doe, cmap='gray', dpi=300)
plt.imsave('doe_binary_print.png', binary_doe, cmap='gray', dpi=300)

print("✓ Saved: doe_grayscale_print.png and doe_binary_print.png")
print("Print the grayscale version first — it usually gives the best results.")
```

Listing 2: DOE pattern generator (`prototype/doe_pattern.py`). Gerchberg–Saxton algorithm producing a printable cross-beam phase mask. Output: `doe_grayscale_print.png` and `doe_binary_print.png`.